

From Keywords to Context:

Domain-Specific Retrieval-Augmented Generation as the Next Evolution of Enterprise Knowledge Management

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Abstract

Knowledge management (KM) has long struggled with a recurring problem: organizations accumulate large volumes of valuable information, yet workers fail to locate the right piece at the moment of need. Traditional electronic knowledge repositories (EKRs) depend on keyword retrieval, which is well documented to underperform in recall and precision (Blair, 2002; Corral et al., 2005). Retrieval-Augmented Generation (RAG) is a recent artificial intelligence architecture that pairs a generative large language model with an external, curated knowledge base, retrieving relevant passages at query time and using them to ground the model's response. This paper argues that domain-specific RAG, built on curated and authoritative corpora rather than open-internet data, represents a credible next evolution of enterprise KM infrastructure. Drawing on classical KM theory (Nonaka, 1994; Davenport & Prusak, 2000; Alavi & Leidner, 2001), the well-known limitations of keyword retrieval (Blair, 2002; Corral et al., 2005), and recent peer-reviewed evidence from biomedical RAG implementations (Ke et al., 2025; Amugongo et al., 2024), the analysis shows that domain-specific RAG addresses three persistent KM challenges: hallucination, retrieval failure, and institutional knowledge loss. Organizational subculture (Ravishankar et al., 2011) and human values (Tams et al., 2020) remain the dominant predictors of whether such a system will be adopted.

Keywords*:* knowledge management, retrieval-augmented generation, electronic knowledge repositories, large language models, domain-specific corpora, hallucination, organizational learning

Introduction

The volume of information held within modern organizations has grown faster than the human capacity to retrieve it. Knowledge workers spend a significant portion of every workweek searching for, recreating, or simply failing to find information that exists somewhere inside their own organization (Davenport & Prusak, 2000). The discipline of knowledge management (KM) emerged in response to this problem, proposing tools, processes, and organizational practices to capture, store, and disseminate knowledge effectively (Dalkir, 2017). Yet despite decades of investment in electronic knowledge repositories (EKRs), document management systems, intranets, wikis, and shared drives, organizations continue to underutilize them (Tams et al., 2020).

Two related problems explain the gap between KM aspiration and KM reality. First, retrieval mechanisms inside EKRs have remained largely unchanged: keyword search, with all its documented limitations, still dominates (Blair, 2002; Corral et al., 2005). Second, much of what makes organizational knowledge valuable is tacit whereby it is embedded in expertise, judgment, and shared practice Tacit knowledge resists capture in any document repository (Nonaka, 1994; Blair, 2002).

Recent advances in artificial intelligence, specifically the development of Retrieval-Augmented Generation (RAG), introduce a new technical possibility for KM. A RAG system combines a generative large language model (LLM) with an external retrieval component: rather than relying solely on what the model learned during pre-training, RAG retrieves relevant documents from a designated corpus at the moment of a user query and supplies those documents as context for the model's generation (Lewis et al., 2020; Gao et al., 2023). When the corpus is curated, vetted, and domain-specific, for example, PubMed for medical questions or a

firm's internal policy library for corporate questions, the resulting system produces grounded, traceable, and conversationally accessible responses (Amugongo et al., 2024).

This paper argues that domain-specific RAG, configured as enterprise knowledge infrastructure, advances the practice of KM in three concrete ways: it reduces hallucination through grounding in vetted sources, it overcomes the keyword-recall limitations identified decades ago, and it lowers the cost of knowledge transfer when expert employees leave. The argument proceeds in four parts. The literature review establishes the theoretical foundation and documents the limitations of keyword retrieval. The body section maps RAG to KM theory, explains why corpus specificity matters, and reviews recent empirical evidence. The discussion examines organizational prerequisites, drawing on the literature on subculture and human values in KM (Ravishankar et al., 2011; Tams et al., 2020). The conclusion identifies limitations and avenues for further research.

Literature Review

KM Theory and the Tacit–Explicit Distinction

KM as a formal discipline emerged in the 1990s. Nonaka (1994) introduced the Socialization-Externalization-Combination-Internalization (SECI) model, which frames knowledge creation as a continuous conversion between tacit and explicit forms. Tacit knowledge which is the unspoken know-how of skilled practitioner is hard to codify, while explicit knowledge can be written down. Davenport and Prusak (2000) defined knowledge as a fluid mix of framed experience, values, contextual information, and expert insight, noting that organizational knowledge becomes embedded not only in documents or repositories but also in routines, processes, practices, and norms. Alavi and Leidner (2001) provided the foundational

review that mapped KM systems to organizational outcomes and identified human and cultural factors as central to KM success.

Three observations from this literature shape the present argument. First, the goal of EKRs is to externalize tacit knowledge so it can be combined and re-internalized by others (Nonaka, 1994). Second, an EKR's value depends not on the documents it contains but on whether the right person can locate and apply the right document at the right moment (Davenport & Prusak, 2000; Dalkir, 2017). Third, organizational culture, particularly the willingness of experts to share what they know, mediates everything (Ravishankar et al., 2011; Tams et al., 2020).

Dalkir (2017) consolidated the KM literature into an integrated knowledge management cycle that includes knowledge capture or creation, knowledge sharing and dissemination, and knowledge acquisition and application. Each stage in this cycle imposes a different requirement on supporting technology. Capture requires tools that lower the friction of documentation. Sharing and dissemination require retrieval mechanisms that surface the right material when a user needs it. Acquisition and application require interfaces that present knowledge in a form a worker can act on. The persistent weakness of traditional EKRs has been the second stage: even when documents are captured and stored, retrieval failure prevents the cycle from closing (Blair, 2002; Dalkir, 2017). RAG, by improving retrieval and synthesis, addresses the bottleneck stage.

Limitations of Keyword Retrieval

Blair (2002) asked whether knowledge management was hype, hope, or help, and identified retrieval as one of the field's weakest links. He observed that keyword searches based on the literal content of documents place an unrealistic burden on the searcher, who must anticipate, in advance, the words and phrases that will appear in the documents of interest. Blair

and Maron (1985) had shown empirically that this anticipation is a skill humans perform poorly: a controlled study of a full-text legal retrieval system found that lawyers retrieved only about 20% of the documents relevant to their queries while believing they had retrieved most of them.

Corral, LaBrie, and St. Louis (2005) extended this analysis. They documented that keyword retrieval suffers from spelling inconsistencies, abbreviation variance, term ambiguity, redundancy, and the Zipf problem: as the document collection grows, the number of documents containing any given common term grows quickly, swelling result sets with irrelevant matches. Their proposed remedy; dynamic visual keyword hierarchies and context Lense leveraged the recognition versus recall distinction from cognitive psychology to make retrieval more effective. While dynamic hierarchies addressed part of the problem, they did not solve the underlying issue: keyword retrieval, whether flat or hierarchical, does not understand what the user is actually asking.

Retrieval-Augmented Generation

RAG was introduced by Lewis et al. (2020) as a method that combines the linguistic fluency of LLMs with grounding in external documents. In a RAG system, a user query is first converted to a vector embedding; semantically similar passages are retrieved from a vector database; and those passages are provided to the LLM as context for generating a response. Unlike keyword search, RAG retrieval is semantic: the user can phrase a question in any natural way, and the system will retrieve passages that mean roughly the same thing even when no words overlap. Unlike a standalone LLM, RAG can cite sources, allowing users to verify the system's claims.

A growing body of research has documented the effectiveness of RAG in addressing the long-standing limitations of LLMs themselves. Gao et al. (2023) provided a comprehensive

survey of the field and identified hallucination reduction as RAG's principal benefit: by constraining the LLM's generation to retrieved evidence, RAG reduces the model's tendency to fabricate plausible sounding but false content. Recent peer-reviewed work has begun to quantify these benefits in specialized domains. Ke et al. (2025), in a systematic review and meta-analysis published in the Journal of the American Medical Informatics Association, examined twenty studies comparing baseline LLM performance against RAG-enhanced LLM performance in biomedical question-answering and reported an overall improvement in odds ratio of 1.35 in favor of RAG. Amugongo et al. (2024), in a PLOS Digital Health review, found that grounding LLM generation in clinical guidelines, peer-reviewed literature, and patient records consistently improved accuracy, contextual relevance, and traceability across multiple deployments.

RAG as Next-Generation Knowledge Management

Mapping RAG to KM Theory

RAG can be reframed in classical KM terms. The vector database serves as a repository of externalized explicit knowledge; the Combination step in Nonaka's (1994) SECI model. The semantic retriever performs a function analogous to a knowledgeable librarian who understands the user's intent, identifies relevant material, and surfaces it. The generative LLM combines the retrieved material into a single conversational response, supporting the Internalization step by presenting knowledge in a form the user can readily absorb and act on (Dalkir, 2017).

Several features distinguish a RAG-based KM system from a traditional EKR. First, retrieval is semantic rather than lexical, removing the keyword-anticipation burden documented by Blair (2002) and Corral et al. (2005). Second, the response is synthesized rather than served as a list of documents, meaning the user does not have to scan five files to assemble an answer. Third, citations are included, so the user can verify the source and pursue depth if needed.

Fourth, the corpus can be tightly controlled by the organization. This last property is the property that distinguishes domain-specific RAG from a general-purpose chatbot.

These features map directly onto the recognition-versus-recall distinction that Corral et al. (2005) drew from cognitive psychology. Keyword search requires recall: the user must remember or reconstruct the precise vocabulary used in documents. RAG requires recognition: the user describes the problem in natural language, the system surfaces relevant passages, and the user recognizes which parts answer the question. Because recognition consistently outperforms recall as a retrieval strategy (Corral et al., 2005), the cognitive ergonomics of RAG are favorable. The same insight that motivated visual hierarchies in 2005 motivates RAG in 2026, but RAG extends the principle by also synthesizing the recognized material into a usable response.

Why Corpus Specificity Matters

A general-purpose LLM trained on the open internet has learned from a mixture of vetted and unvetted sources. When such a model is asked a domain-specific question, say, the recommended first-line treatment for a particular cancer, it may produce a confident-sounding response drawn from outdated, contested, or simply wrong information. A domain-specific RAG system, by contrast, retrieves only from a curated authoritative corpus: PubMed for medicine, court records for law, the firm's policy manuals for corporate decision-making. The model's response is then constrained to the retrieved evidence.

Ke et al. (2025) and Amugongo et al. (2024) both observe that the curation of the underlying corpus is the most consequential design decision in a domain-specific RAG system. A second decision concerns the granularity of retrieval: how passages are chunked, embedded, and indexed determines retrieval quality (Gao et al., 2023). Kumar et al. (2026), in a systematic literature review of enterprise RAG and document automation, concluded that the transition from

generic LLM applications to grounded, domain-specific architectures is the central trend in enterprise AI, and that this transition directly addresses LLM limitations including hallucination, static knowledge cutoff, and inability to incorporate proprietary information.

A further consideration is access control. Enterprise corporations often contain sensitive material such as contracts, customer records and internal strategy documents that cannot be exposed to every employee. A domain-specific RAG system makes it possible to enforce document-level permissions at the retrieval stage, ensuring that the LLM never sees passages the requesting user is not authorized to read. This property maps cleanly onto Dalkir's (2017) governance dimension of KM and addresses one of the recurring concerns in EKR adoption: workers may withhold valuable knowledge from a shared repository because they cannot control who sees it (Tams et al., 2020). A RAG architecture with role-aware retrieval reduces that.

Table 1 summarizes the principal differences between traditional keyword-based EKR retrieval and domain-specific RAG retrieval across seven dimensions.

Table 1

Comparison of Traditional Keyword EKR and Domain-Specific RAG

Dimension	**Traditional Keyword EKR**	**Domain-Specific RAG**
Retrieval mechanism	Lexical (term match)	Semantic (vector similarity)
User burden	Anticipate document terminology	Express intent in natural language
Output form	Ranked list of documents	Synthesized response with citations
Sensitivity to spelling	High	Low
Primary failure mode	Too few or too many results	Hallucination when corpus is sparse
Role of corpus quality	Affects findability	Determines factual correctness
SECI stage supported	Combination	Combination and Internalization

Note. Adapted from Blair (2002), Corral et al. (2005), Lewis et al. (2020), Gao et al. (2023), and Nonaka (1994).

Evidence from Biomedicine and Enterprise Deployments

Biomedicine has been the leading domain for RAG research, in part because the consequences of hallucination are immediate and high-stakes. Wu et al. (2024) developed Medical Graph RAG, a system that retrieves from curated medical knowledge sources at inference time and reported substantial improvements in medical question-answering accuracy compared with baseline LLMs. Yang et al. (2024) demonstrated that integrating clinical guidelines as the retrieval source for an LLM-based system improved both accuracy and contextual relevance of medical responses. The meta-analysis by Ke et al. (2025) consolidates these findings across twenty studies and confirms a measurable improvement in performance attributable to RAG.

Outside biomedicine, enterprise deployments have shown comparable benefits in document-heavy domains. Kumar et al. (2026) cataloged enterprise applications including legal research, financial advisory document retrieval, and technical support, finding consistent reductions in hallucination and improvements in response traceability when authoritative corpora are used. A practical antecedent exists in financial services: an internally developed semantic retrieval system was deployed at a banking institution to give employees natural-language access to product documentation, replacing the previous practice of scanning long product catalogues. The system was adopted and scaled across the institution. While practitioner deployments of this kind do not yet constitute peer-reviewed evidence, they illustrate the lab-to-market gap identified by Kumar et al. (2026): academic RAG prototypes outnumber documented production deployments, and case studies of successful deployments at the enterprise scale remain rare.

The pattern across biomedicine, finance, and law is consistent: the more an industry depends on accurate retrieval from a defined body of authoritative documents, the larger the benefit of switching from keyword-based to semantic, generation-augmented retrieval.

Discussion: Organizational Prerequisites and Limitations

The most consequential observation in KM literature is that technology alone is insufficient. Ravishankar, Pan, and Leidner (2011), in a multilevel case study of a KMS implementation at an Indian information technology services firm, demonstrated that subunit-level subcultures determined the success or failure of the system far more than technical alignment did. Three subcultures emerged in their data: an enhancing subculture that mirrored the broader organizational values and adopted the KMS readily; a chameleon subculture whose adoption fluctuated based on the team to which members were temporarily assigned; and a countercultural subculture that rejected the KMS in favor of identification with an external client. The same technical system, with the same capabilities, succeeded in one subunit and failed in another. Tams et al. (2020), in a large-scale survey of 233 knowledge workers, found that human values, specifically benevolence, universalism, and self-direction predicted the post adoptive use of EKRs. Workers whose values aligned with sharing and openness were more likely to contribute to and innovate with an EKR, while workers whose values emphasized self-enhancement or conservation were less likely. The implication is direct: the most technically sophisticated RAG system will fail if its intended users do not contribute to its corpus or trust its outputs.

Two further limitations bear on any RAG-based KM system. First, tacit knowledge remains outside the reach of document-based architecture. Blair (2002), drawing on Dreyfus and Dreyfus (1986), argued that most expertise cannot be reduced to rules or text, and that genuine

transfer of expertise requires human-to-human interaction within communities of practice. RAG can disseminate explicit knowledge effectively, but it cannot substitute for apprenticeship in tacit domains. Second, the curation of the corpus is itself an expert activity that requires ongoing investment. A RAG system is only as good as its corpus, and corpus quality decays without active maintenance; a long-standing observed concern in the KM literature (Davenport & Prusak, 2000; Dalkir, 2017).

Implementation choices outside the technology itself also matter. Dalkir (2017) frames KM implementation through the interaction of business, technology, organization, people, and process dimensions as a perspective consistent with the broader information systems literature on socio-technical alignment. For RAG specifically, the technology dimension is the easiest to address. Open-source frameworks, managed vector databases, and commercial LLM application programming interfaces are now widely available. The harder dimensions are organizational and procedural. Who owns the corpus? Who decides what is authoritative? How are sensitive documents partitioned across user roles? Who is accountable when a RAG response is wrong? These questions echo those that surrounded the introduction of EKRs twenty years ago (Alavi & Leidner, 2001), and the experience of that earlier wave of implementations suggests that organizations that treat RAG as primarily a technology project will struggle, while those that treat it as a knowledge governance project will succeed.

Conclusion

This paper has argued that domain-specific Retrieval-Augmented Generation represents a credible next evolution of enterprise knowledge management infrastructure. Three mechanisms have been identified. First, semantic retrieval addresses the long-standing keyword-search limitations documented by Blair (2002) and Corral et al. (2005). Second, grounding LLM

generation in a curated, authoritative corpus reduces hallucination, as quantified in the biomedical literature (Ke et al., 2025; Amugongo et al., 2024). Third, by capturing institutional knowledge in a query able form, RAG-based systems can mitigate the loss that occurs when expert employees leave the organization (Davenport & Prusak, 2000).

The argument is bounded by clear limitations. RAG systems cannot capture tacit knowledge, and they depend critically on the quality of the underlying corpus. The most carefully engineered RAG system will also fail if the surrounding organizational culture does not support contribution and use, as the subculture and human-values literatures have shown (Ravishankar et al., 2011; Tams et al., 2020). The lab-to-market gap identified by Kumar et al. (2026) further constrains what can currently be claimed: peer-reviewed evidence of enterprise RAG deployments is still thin compared to academic prototypes.

Future research could pursue four directions. First, real-world studies of enterprise RAG rollouts would help measure their actual cost and value at scale. Second, the interaction between RAG and the subcultures identified by Ravishankar et al. (2011) is worth studying. Does semantic retrieval ease subunit resistance, or make it worse? Third, hybrid systems that combine vector retrieval with knowledge graphs (Edge et al., 2024) are worth exploring in a KM setting. Fourth, the values framework from Tams et al. (2020) could be applied to RAG adoption to see which workers take it up. At its core, KM is about getting the right information to the right person at the right moment. RAG handles the technical side well; the human side is still up to us.

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Appendix A

Generative AI Prompts Used in This Research

Claude (Anthropic) was used as research and writing assistant throughout the development of this paper. The tool was used for literature triangulation, source verification, structural feedback, paraphrasing, and humanization of overly formal phrasing. All citations were independently verified against the original source material, and all final wording reflects the author's own voice and judgment. The prompts below are reproduced in the order they were used during the research process.

Prompt 1: Topic Validation

> "*I have a knowledge management research idea. Without blindly agreeing with me, tell me if it's worth exploring. Pros and cons. The idea: a domain-specific RAG system trained on curated, authoritative data instead of the whole internet (e.g., a medical LLM trained on PubMed rather than general web data). Could reduce hallucinations, scale to corporate settings. Does this fall under knowledge management or am I stretching the definition? Connect it to actual KM theory if so.*"

Use: Output used to confirm the topic was theoretically grounded in KM literature before committing to the proposal. Suggested directions (SECI model, Nonaka, Davenport & Prusak) were independently verified.

Prompt 2: Literature Search

> "*Scour the internet, especially peer-reviewed journals and published research for sources on the connection between RAG technology and knowledge management. Include systematic reviews if any exist. I need real academic sources, not blog posts or vendor marketing. Share the links so I can verify.*"

Use: Output produced an initial source list including the MDPI systematic review, the JAMIA biomedical meta-analysis, and the PLOS Digital Health healthcare review. Each source was independently retrieved and read before citation.

Prompt 3: Paper Structure Planning

> "*I need to write a 3000-word KM research paper on domain-specific RAG. APA format. Needs cover page, abstract with keywords, introduction with thesis, literature review, body, conclusion with limitations and future research, and 10+ references from quality journals. Outline a structure that fits.*"

Use: Output informed the section structure (Introduction → Literature Review → RAG as Next-Generation KM → Discussion → Conclusion). Word allocation per section was adjusted during drafting.

Prompt 4: Source Synthesis from Uploaded PDFs

> "*Uploaded: Blair (2002), Corral et al. (2005), Ravishankar et al. (2011), Tams et al. (2020). For each, pull out the strongest arguments that connect to my RAG thesis. I want to know which paper supports which part of the argument so I cite accurately.*"

Use: Output used to map each source to a specific section of the paper. Direct quotes and statistics were verified against the original PDFs before inclusion.

Prompt 5: Comparison Table

> "*Generate a comparison table for my paper: traditional keyword EKR search vs. domain-specific RAG. Cover retrieval mechanism, user burden, output form, failure mode, role of corpus quality, and SECI stage supported. Keep it clean.*"

Use: Output became Table 1 in the paper. Citations added by the author, and the SECI stage row was adjusted to better reflect Nonaka's original framing.

Prompt 6: Future Research Paragraph

> "*Paraphrase this paragraph in simpler, straightforward English, about 120 words. [Draft of future research directions attached]*"

Use: Output used to refine the closing paragraph of the conclusion section. Final wording was further tightened by the author.

Prompt 7: One-Slide Research Summary

> "*Do one PowerPoint slide summarizing my research. Should carry the thesis without being text heavy. Visual hierarchy matters. Don't make it look like a generic corporate template.*"

Use: Output produced the single-slide research brief used for in-class presentation. Content reflects the paper's thesis.

Appendix B

List of Abbreviations

The following abbreviations are used throughout this paper. They are listed alphabetically for ease of reference.

Abbreviation	Meaning
AI	Artificial Intelligence
EKR	Electronic Knowledge Repository
FAISS	Facebook AI Similarity Search (an open-source vector database library)
JAMIA	Journal of the American Medical Informatics Association
KM	Knowledge Management
KMS	Knowledge Management System
LLM	Large Language Model
PubMed	U.S. National Library of Medicine biomedical research database
RAG	Retrieval-Augmented Generation
SECI	Socialization, Externalization, Combination, Internalization (Nonaka, 1994)

Note. Names of proprietary tools and journals are listed for clarity but are not strictly abbreviations.